

Energy and Enthalpy

Friday, 28 August 2020

10:47 AM

Introduction

- In a reaction, reactant bonds must break and product bonds must form
 - Breaking bonds is endothermic (takes energy from surroundings)
 - Forming bonds is exothermic (releases energy into surroundings)
- **Activation Energy:** Minimum amount of energy required to break reactant bonds for reaction to occur
- **System:** Chemicals taking part in the reaction and their bonds
- **Surroundings:** The whole universe surrounding it
- **Energy (J):** The ability to do work
- **Law of Conservation of Energy:** Energy cannot be created nor destroyed, but may change form

Enthalpy

- **Enthalpy (H):** Total internal energy of a system, includes chemical potential energy + kinetic energy of particles
 - Enthalpy cannot be measured - we measure energy change to calculate enthalpy
 - Always measured at **standard conditions**
 - 25°C/298°K & 100kPa/1atm
- **ΔH** (Change in enthalpy) = $H_{(\text{products})} - H_{(\text{reactants})}$
 - $1\text{kJ g}^{-1} = 1000\text{MJ tonne}^{-1}$

Heat of Combustion (ΔH_c)

- **ΔH_c (J/kJ mol⁻¹):** Enthalpy change when 1 mole of a compound reacts completely with oxygen (standard states and conditions)
- $\Delta H_c = \frac{\Delta H}{n(\text{fuel})}$
- Higher molar mass fuels are more efficient (higher ΔH_c)

Experiment to Calculate ΔH_c

- $\Delta H(J) = mc\Delta T$
 - m = mass of water heated (g)
 - c = specific heat capacity of water (4.18J g⁻¹ K⁻¹ OR 4.18J g⁻¹ °C⁻¹)
 - ΔT = temperature change in water (°K / °C)

Problems

- Not all heat energy goes towards heating up the water
- Not complete combustion (soot formed)
- Delay between taking cap off and starting flame where fuel is evaporating

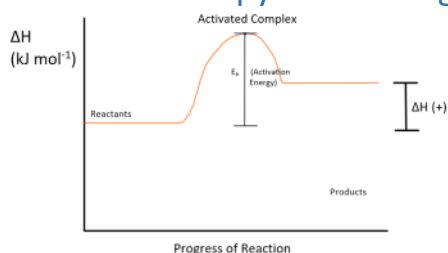
Improvements

- Insulating shield
- Constant air flow so complete combustion is more likely

Exothermic vs Endothermic

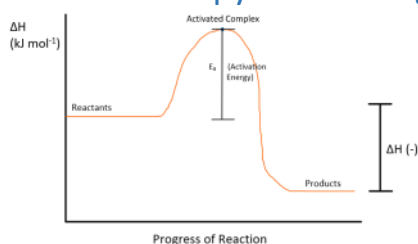
Feature	Endothermic	Exothermic
ΔH	Positive	Negative
Product Bond Strength (compared to reactants)	Weaker	Stronger
Energy Flow	$E_k \rightarrow \text{CPE}$	$\text{CPE} \rightarrow E_k$
Surrounding Temperature	Decreases	Increases
Heat/Energy in Equation	Reactants	Products

Endothermic Enthalpy Profile Diagram



- E.g.¹ $\text{CaCO}_3 + \text{Heat} \rightarrow \text{CaO} + \text{CO}_2$

Exothermic Enthalpy Profile Diagram



- E.g.¹ $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{Heat}$
- E.g.² $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3 + \text{Heat}$
 - Very high activation energy as nitrogen has triple bonds
- **Activated Complex:** Unstable arrangement of atoms in which reactant bonds are breaking and product bonds are forming

Catalysts

- **Catalyst:** A chemical which increases the rate of reaction by providing an **alternative pathway with a lower activation energy**
- DOES NOT LOWER ACTIVATION ENERGY OF REACTANTS - it is a fixed value
- Homogeneous Catalyst: Same phase as all reactants and products
 - Takes part in reactions
- Heterogeneous Catalyst: Different phase to reactants/products
 - Provides a common adsorption surface for reactions to occur

Fuels

- **Renewable:** Obtained from natural resources that can be constantly replenished
- **Non-Renewable:** Used faster than they can be replaced

Fossil Fuels

- Contain chemical energy from photosynthesis millions of years ago
- **Crude Oil:** A mixture of hydrocarbons formed from geological action on decayed matter over millions of years
 - Not useful naturally, but can be separated into many important fuels by fractional distillation

Biofuels

- Biofuel: A fuel produced from biomass
- Renewable
- **Carbon Neutral:** CO_2 emitted by burning was taken from the atmosphere during photosynthesis
 - Still has carbon emissions as other fuels are required to create biofuels (fertilizer, transportation, manufacturing)
 - Trees are often cut down to make space for biofuel crops, reducing the amount of CO_2 taken from the atmosphere

Bioethanol

Formation

- Fermentation of crops such as wheat, maize, sugar beet and barley

- ◆ Fermentation: Enzymes breaking down glucose into ethanol
- Hydration of ethene at high temperature (800°C) and high pressure
 - ◆ $\text{CH}_2\text{CH}_2 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$

Usage

- Used in vehicles to reduce net carbon output into the atmosphere
- E₁₀ (90% petrol 10% bioethanol) will work in regular engines
- E₈₅ (15% petrol 85% bioethanol) requires modified engines

Biodiesel

- **Biodiesel:** Mixture of methyl esters produced by a chemical reaction between vegetable oils or animal fats and methanol

Formation (transesterification)

- Made from: Cooking oil, soybean oil, animal fats
- Catalysts: KOH/NaOH

Usage

- Can be used in diesel engines without modification
- Transportation, heat, charging electronics

Ideal Gases

Kinetic Theory

1. Particles move in continuous random linear motion
2. Negligible attraction/repulsion between particles
 - No intermolecular forces
3. Collisions between particles are elastic - no E_k loss
4. Average kinetic energy of particles is proportional to temperature
 - $E_K = \frac{1}{2}mv^2$
5. Particles are spherical with negligible volume

Gas Properties

- **Takes shape and volume of container:** Random motion; Particles spread out as far as they can
- **Easily compressed:** Volume of particles negligible; Free space to compress them
- **Diffuse through other gases:** Random motion + negligible attraction; Can spread out between molecules of other gases
- **Exert pressure:** Random motion; Collide with container walls

Real vs Ideal Gases

- Gases that behave according to the kinetic theory are called **ideal gases**
- When relating gases to ideal gases, we assume high temperature and low pressure as it means there will be less attraction between particles
- Real gases will always have attraction between particles, and particles have volume

Gas Laws

- **Boyle's Law:** $PV = k$
 - Pressure and volume are inversely proportional
 - Assumes constant temperature and moles
- **Charles's Law:** $\frac{V}{T} = k$
 - Volume and temperature are directly proportional
 - Assumes constant pressure and moles
- **Avogadro's Law:** $\frac{V}{n} = k$
 - Volume and moles are directly proportional
 - Assumes constant pressure and temperature
- **Ideal Gas Equation:** $PV = nRT$
 - P : Pressure (kPa)
 - V : Volume (L)
 - n : Amount of gas (mol)
 - R : Gas constant (8.341)

- T : Temperature ($^{\circ}\text{K}$)

Temperature and Phase Change

- $\uparrow \text{Temp} = \uparrow E_k$
- With enough E_k , particles can break apart and change state
- When in the process of melting/boiling, temperature and E_k remain constant

Units of Measurement

- $\text{Pressure} = \frac{\text{Force}}{\text{Area}}$
 - $\text{Pascal(Pa)} = \frac{\text{Newtons}}{\text{m}^2}$
 - $1\text{atm} \approx 100\text{kPa}$
- Absolute zero: Particles have no kinetic energy
 - $0^{\circ}\text{K} = -273.15^{\circ}\text{C}$

Class

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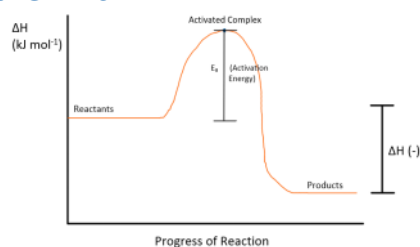
Introduction

- To react, reactant bonds must break and product bonds must form
 - Breaking bonds is endothermic (takes energy from surroundings)
 - Forming bonds is exothermic (releases energy into surroundings)
- **System:** Chemicals taking part in the reaction and their bonds
 - A, B and AB
- **Surroundings:** The whole universe surrounding it
 - E.g. In a solution: Most to least immediate - water, beaker, air, etc.
- **Activation Energy:** Minimum amount of energy required for reaction to occur

Exothermic vs Endothermic

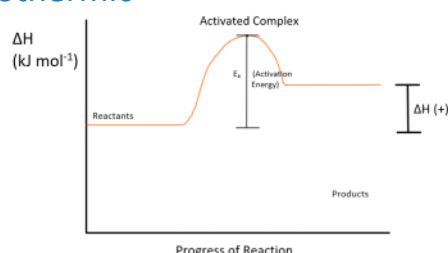
Feature	Endothermic	Exothermic
ΔH	Positive	Negative
Product Bond Strength (compared to reactants)	Weaker	Stronger
Chemical Potential Energy	Higher	Lower
Surrounding Temperature	Decreases	Increases
Heat/Energy	Reactants	Products

Exothermic



- E.g.¹ $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{Heat}$
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Endothermic



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Catalysts

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- DOES NOT LOWER ACTIVATION ENERGY OF REACTANTS - it is a fixed value

Enthalpy

- **Enthalpy (H):** Heat energy at constant pressure
 - Enthalpy cannot be measured - we measure energy change to calculate enthalpy

- **ΔH** : Change in enthalpy

- $\Delta H = H_{(\text{products})} - H_{(\text{reactants})}$

Enthalpy Change of Combustion (ΔH_c)

- **Enthalpy Change of Combustion (ΔH_c)**: Enthalpy change when 1 mole of a compound reacts completely with oxygen (standard states and conditions)
 - Standard Conditions: 25°C & 1atm OR 298°K & 100kPa
- $\Delta H(J) = mc\Delta T$
 - m = mass of water heated (g)
 - c = specific heat capacity of water (4.18J g⁻¹ K⁻¹ OR 4.18J g⁻¹ °C⁻¹)
 - ΔT = temperature change in water (°K / °C)
- $\Delta H_c = \frac{\Delta H}{n(\text{fuel})}$
- Higher molar mass fuels are more efficient (higher ΔH_c)

Experiment to Calculate ΔH_c

- Metal tin of water

Problems

- Not all heat energy goes towards heating up the water
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Improvements

- Insulating shield
- Constant air flow so complete combustion is more likely

Biofuels

- Biofuel: A fuel produced from biomass rather than other resources, such as fossil fuels
 - Renewable (organic matter can be grown within a lifetime, fossil fuels take millions of years to form)
 - Carbon-neutral

Carbon Neutral

- No net carbon released into the atmosphere
- Carbon dioxide released by fuels into the atmosphere are taken from the atmosphere by the plants in the first place
- However, other fuels are required to create biofuels, so it is not completely carbon-neutral
 - Fertiliser, transportation, manufacturing
- Trees are often cut down to make space for biofuel crops, reducing the amount of CO₂ absorbed from the atmosphere

Bioethanol

Formation

- Sugar fermentation
 - Fermentation: Biomass decomposed using bacteria/enzymes
 - ◆ Sugar comes from crops such as wheat, maize, sugar beet and barley
- Reacting ethylene (ethene) with steam at 750-850°C for a short period of time (< 0.5s)
 - $\text{CH}_2\text{CH}_2 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$

Usage

- Used in vehicles to reduce net carbon output into the atmosphere
- E₁₀ (90% petrol 10% bioethanol) will work in regular engines
- E₈₅ (15% petrol 85% bioethanol) requires modified engines

Biodiesel

- Definition: A fuel composed of mono-alkyl esters of long chain fatty acids derived from vegetable oils or animal fats

Formation

- Process called **transesterification**

- KOH or NaOH used as catalysts
- Methoxide reacts with a triglyceride (from cooking oil, soybean oil, animal fats)
- Glycerine is stripped from the oil
- Forms methyl esters (biodiesel) and glycerine

Usage

- Can be used in diesel engines without modification
- Transportation, heat, charging electronics

Ideal Gases

Crash Course

- Lower volume = Higher pressure
- Boyle's Law: $PV = k$
 - Pressure x Volume = Some Constant Value
 - The constant differs for different gases and at different temperatures
- $\frac{V}{T} = k$
- $\frac{V}{n} = k$
- Ideal Gas Law: $PV = nRT$
 - P : Pressure
 - Pascal: Newtons per metre squared
 - 1atm = 100kPa
 - V : Volume
 - n : Moles
 - R : Universal gas constant
 - T : Temperature - Kinetic energy of particles
 - Lower temperatures mean lower pressure as particles are bumping into the container walls less
- STP: 0°C and 100kPa
- Absolute zero: All movement of all particles stops
 - 0°K or -273.15°C

Class

- $PV = nRT$
 - P : Pressure (kPa)
 - V : Volume (L)
 - n : Amount of gas (mol)
 - R : Gas constant (8.341 - on data sheet)
 - T : Temperature (°K)

Assumptions

- Particles continuously move in random straight lines
 - Gas particles are spheres
 - Pressure is the force exerted by particles on the walls of the container
 - Collisions between particles are elastic - no loss of E_k during collisions
 - No intermolecular forces between particles
 - Temperature of a gas is proportional to average kinetic energy of particles
 - Volume occupied by particles is negligible
- When talking about ideal gases, we assume high temperature and low pressure as it means there will be less attraction between particles
 - Using $E_k = \frac{1}{2}mv^2$, we can determine that lighter molecules will move faster than heavier molecules

Chapter 1 - A Particle View of Matter

1.2 - Kinetic Theory of Gases

- All gases take the shape of their container, exert pressure, have a low density, can be compressed and readily diffuse through other gases
- Kinetic theory is a model used to describe this behaviour
 - Gases are composed of particles that move in continuous random motion
 - Attraction and repulsion of particles is negligible
 - Particles widely spaced, volume of particles is negligible compared to volume of gas
 - Kinetic Energy (E_K) = $\frac{1}{2}mv^2$
 - Speed = v, mass = m
 - Average E_K of a particle is proportional to its temperature and is the same for all gases at the same temperature
 - Particles do not lose E_K due to collisions with other particles

1.3 - Gas Behaviour (Applying 1.2)

- **Shape of Container and Low Density ($d = m/V$):** Random motion of particles, negligible attraction; Spread out as far as possible
- **Can be Compressed:** Negligible volume and widely spaced, free room to be compressed
- **Readily Diffuse Through Other Gases:** Random motion, negligible attraction; can spread out between molecules of any other gas
- **Exert Pressure:** Particles move randomly in straight lines; collide with container walls exerting pressure. Increases with frequency and force of collisions

1.4 - Kinetic Theory and Ideal Gases

- Kinetic theory only applies to idealised gases which we call **ideal gases**
- Difference between ideal and real gas:
 - Particles of an ideal gas have no volume, real gases do
 - Particles of an ideal gas have no attraction to each other, real gases do
- Differences only make a difference in extreme situations, eg:
 - At 0°K (-273°C) an ideal gas would have 0 volume, real gas will always have a volume
 - If a real gas is cooled and/or compressed enough it condense into a liquid due to the attraction between particles. Ideal gases have no attraction

1.5 - Kinetic Theory and Condensed Phases of Matter

- Can be extended to explain behaviour of solids and liquids
- Solids: Attraction between particles is very strong; particles packed so tightly they can only vibrate around a point within a structure
- Liquids: Strong attraction between particles, not as strong as solids; Particles can move past each other (fill a container) but are incompressible

1.6 - Temperature, Energy and Phase Change

- Heating a substance will increase the kinetic energy (temperature) of its particles
- With enough E_K , particles can break apart to change state from solid → liquid/liquid → gas
- The temperature at which this happens is the melting point/boiling point
- When a substance is in the process of melting/boiling its temperature and E_K is constant
- Cooling a substance will have the opposite effect

Chapter 13 - Energy Flow and Chemical Change

13.1 - Understanding Energy

- **Energy:** The capacity to do work (joules)
- Various forms of energy are interchangeable (e.g. Heat can be converted into light)
- **Law of Conservation of Energy:** Total amount of energy present is constant

13.2 - Energy Changes in Chemical Reactions

- **Enthalpy (H):** The total energy present in a system
 - Includes chemical potential energy (energy stored in bonds) and kinetic energy of the particles
 - ? □ Strong bond = low chemical potential energy; weak bond = high chemical potential energy
 - Kinetic energy increases with temperature
- During a chemical change bonds are broken and new bonds are formed
 - Chemical potential energy will change
 - $\Delta H = H_{(\text{products})} - H_{(\text{reactants})}$
- If a reaction happens at constant pressure (e.g. Open beaker/test tube) the heat change of the system is equal to ΔH , and ΔH is called the **heat of reaction**
- E.g. $\text{CH}_4 + 2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{CO}_2 + \text{Energy}$
 - Weaker bonds (high CPE) $\rightarrow$ Stronger bonds (low CPE)

13.3 - Endothermic and Exothermic Reactions

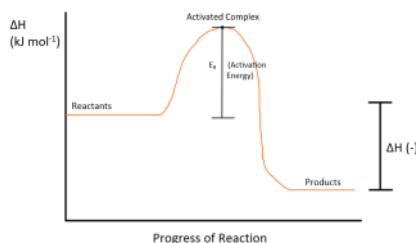
Feature	Endothermic	Exothermic
Energy Flow	Kinetic $\rightarrow$ CPE	CPE $\rightarrow$ Kinetic
Surrounding Temperature (Changes with system)	Decrease	Increase
ΔH	Positive	Negative

13.4 - Representing Energy Changes in Chemical Reactions

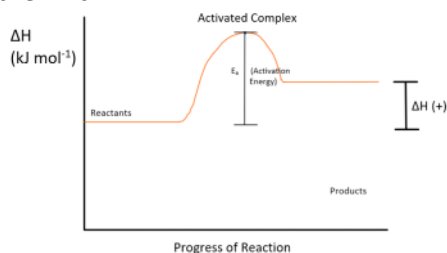
- May be shown using an enthalpy profile diagram (EPD) or a chemical equation
- Chemical Equations
 - E.g.¹ $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O} + 572\text{kJ}$; $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, $\Delta H = -572\text{kJ}$
 - E.g.² $\text{CO}_2 + 394\text{kJ} \rightarrow \text{C} + \text{O}_2$; $\text{CO}_2 \rightarrow \text{C} + \text{O}_2$, $\Delta H = 394\text{kJ}$

Enthalpy Profile Diagram

- Exothermic



- Endothermic



13.5 - Energy Effects of Making and Breaking Bonds

- Breaking bonds is endothermic
- Forming bonds is exothermic
- Breaking/Forming stronger bonds absorbs/releases more energy than weaker bonds

13.6 - Comparing Energy in Physical and Chemical Changes

- Physical changes (e.g. melting/boiling/sublimation) have a significantly lower energy change than chemical changes as only weak bonds (IMFs) are broken as opposed to strong bonds (e.g. Covalent bonds)

13.7 - Energy For You

- Plants convert solar energy into chemical potential energy stored in carbohydrates, oils, and proteins during photosynthesis
 - $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2, \Delta H = 2870\text{kJ}$
- When we eat them, cellular respiration causes the reverse of photosynthesis to occur, releasing energy for our bodies
 - $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}, \Delta H = -2870\text{kJ}$

Chapter 10 - Energy Changes in Chemical Reactions

10.1 - Exothermic and Endothermic Reactions

- Chemical energy: Energy stored in chemical bonds
- $1\text{MJ} = 1000\text{kJ} = 1\,000\,000\text{J}$
- Law of Conservation of Energy: Energy cannot be created nor destroyed, but can change form

Energy Changes During Reactions

- Breaking bonds is endothermic, forming bonds is exothermic (separated particles have more energy)
- Energy gained/lost often in the form of heat, or sometimes movement (kinetic energy)
- **Exothermic Reactions:** Energy released from the system
 - Chemical energy of products is less than chemical energy of reactants
- **Endothermic Reactions:** Energy absorbed
 - Chemical energy of products is more than chemical energy of reactants

Phase Changes

- Physical, not chemical change, but still release/absorb energy
- Solid→Liquid→Gas: Endothermic (requires energy)
- Gas→Liquid→Solid: Exothermic (releases energy)

10.2 - Thermochemical Equations, Energy Profile Diagrams and Enthalpy

- **Enthalpy (H):** Stored chemical energy of a substance
- **Heat of Reaction/Enthalpy Change (ΔH)** = $H_{\text{products}} - H_{\text{reactants}}$
- Exothermic: $\Delta H < 0$
- Endothermic: $\Delta H > 0$
- **Thermochemical equation:** Write in the ΔH value to the right of the equation
 - E.g. $\text{CH}_4 + \text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{CO}_2$, $\Delta H = \text{---}$
- Reversing a chemical reaction changes the sign (+/-) of ΔH but not the magnitude
- **Activation energy:** Energy required to break the bonds of reactants so a reaction can occur
 - Exists for both exothermic and endothermic reactions
- Energy profile diagrams show the change in enthalpy over the course of a reaction

Chapter 11 - Fuels and Introduction to Stoichiometry

11.1 - Types of Fuels

- **Non-renewable Resources:** Used faster than they can be replaced
- **Renewable Resources:** Obtained from natural resources that can be constantly replenished

Fossil Fuels

- Fossil fuels contain chemical energy from photosynthesis millions of years ago
- The lower down coal is, the lower its water content and the more energy it can produce
 - $\text{C}_{(\text{s})} + \text{O}_{2(\text{g})} \rightarrow \text{CO}_{2(\text{g})} + \text{Energy}$
- **Crude Oil (Petroleum):** A mixture of hydrocarbons that are mostly alkanes
 - It is of no use as a fuel, but can be separated by fractional distillation into many important fuels

Natural Gas: Mostly methane, a bit of ethane and propane

- Used in homes for heating and cooking
- Can be found in:
 - ◆ Gas reservoirs

- ◆ Component of crude oil
- ◆ The surface of coal

Biofuels

- Fuels derived from plant material
- Carbon Neutral: Carbon emitted by these fuels were taken from the atmosphere during photosynthesis
 - Still some carbon emissions from manufacture and distribution of biofuels
- Bioethanol is produced by fermentation - enzymes breaking down glucose into ethanol
 - $C_6H_{12}O_6 \rightarrow 2CH_3CH_2OH + CO_2$
 - E₁₀: 90% petrol, 10% bioethanol - may be used in engines without modification
- Biogas released in the breakdown of organic waste by bacteria
- Biodiesel: Mixture of esters produced by a chemical reaction between vegetable oils or animal fats and methanol

11.2 - Combustion Reactions

- Complete combustion (excess oxygen) produces CO₂ and H₂O
- Incomplete combustion (limited oxygen) produces CO and H₂O
- Other pollutants such as SO₂ and NO_x may be formed
 - SO₂/NO_x: Formation of acid rain; respiratory problems
 - CO: Poisonous gas
- Greenhouse effect: Heat trapped in earth's atmosphere by greenhouse gases, which causes temperature to increase at earth's surface
 - Consequences include rising sea levels and more extreme weather (e.g. Droughts and floods)

11.3 - Calculations Involving Fuels

Energy Content of Fuels

- Energy Content: The chemical energy a substance contains
- Heat of Combustion: Enthalpy change when a specified amount of fuel undergoes complete combustion
 - Non-pure fuels (e.g. Wood, coal) cannot be expressed in J/kJ mol⁻¹
 - Practically, not all energy is released as heat energy nor can it all be harnessed
- $$kJ\ g^{-1} = \frac{kJ\ mol^{-1}}{M(fuel)}$$
- $1\ kJ\ g^{-1} = 1000\ MJ\ tonne^{-1}$
- Efficiency of a coal power station is around 30-40%
- Efficiency of a gas-fired power plant (natural gas) is just over 40%

Chapter 14 - Gases

14.1 - Introducing Gases (Kinetic Theory)

- Gas Properties:
 - Low density
 - Easily compressed
 - Take the shape and volume of container
 - Mix rapidly together
- Kinetic theory explains this:
 - Volume of particles in a gas is negligible compared to the volume it occupies
 - Average kinetic energy of particles is proportional to temperature, doesn't matter what element it is
 - Gas particles are in random linear motion, collide with each other and their container wall
 - Forces between particles are negligible
- Gases that behave according to kinetic theory are called **ideal gases**
- At high temperatures and low pressures, gases with low molecular mass behave like

ideal gases

- As pressure/temperature increases/decreases, particles are close together for longer, and forces of attraction become noticeable
- **Diffusion:** Gases in a mixture spreading out to homogeneously fill the container
- $Pressure = \frac{Force}{Area}$
 - Pascal (Pa): $\frac{Newtons}{m^2}$
 - Caused by collisions of gas particles against the wall of the container

Rate of Reaction

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Measuring Reaction Rate

- $\text{Rate} = \frac{\text{Quantity of substance used/produced}}{\text{time}}$
 - Many different units (e.g. mol s^{-1} , g s^{-1} , mL s^{-1})
 - May also measure in terms of pH, colour or transparency change
- To calculate, graph the amount of substance used/produced over time
 - Rate = gradient

Collision Theory

- Collision theory states that for a reaction to be successful:
 - Particles must collide with collision energy $\geq$ activation energy (E_a)
 - Particles must collide with the correct orientation
- **Activation energy (E_a):** Minimum amount of energy required to break reactant bonds for a reaction to occur
 - Higher E_a = Slower reaction rate
- If all conditions are met, particles will form a transition state/activated complex
 - **Activated Complex:** Unstable arrangement of atoms with maximum potential energy in which reactant bonds are breaking and new bonds are forming

Maxwell-Boltzmann Distribution

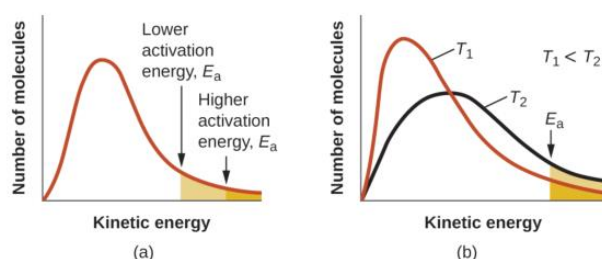


Figure 4. (a) As the activation energy of a reaction decreases, the number of molecules with at least this much energy increases, as shown by the shaded areas. (b) At a higher temperature, T_2 , more molecules have kinetic energies greater than E_a , as shown by the yellow shaded area.

- Area under curve = total number of particles; Always the same
- No particles have no energy
- No maximum energy (should never touch the x-axis to the right)
- Only particles to the right of the activation energy have enough energy to react

Factors Affecting the Reaction Rate

- Increasing rate can be achieved by:
 - Increasing temperature
 - Increasing concentration (solutions)/pressure (gases)
 - Increasing surface area (solids)
 - Adding a catalyst
- **↑Temperature = ↑ E_k = ↑Collision frequency + ↑Collisions with energy $\geq E_a$ = ↑Successful Collisions**
 - Reaction rate doubles for a 10°K/C rise in temperature
- **↑Concentration (solutions)/↑Pressure (gases) = ↑Particles in given space (closer together) = ↑Collision frequency = ↑Successful collisions**
- **↑Surface area (solids) = ↑Area for collisions to occur = ↑Collision frequency = ↑Successful collisions**
 - Heterogeneous reactions can only occur at the surface of reactants
- **Catalyst:** Provides reaction pathway with lower E_a

Catalysts

- **Catalyst:** A chemical which increases the rate of reaction by providing an alternate reaction pathway with a lower activation energy
 - No net consumption
 - Show using a dotted line in enthalpy profile diagrams

- Shifts E_A left in Maxwell-Boltzmann Distribution
- Greater proportion of collisions have energy $\geq E_a \rightarrow$ More successful collisions
- Used to increase rates as it is cheaper than increasing temperature/pressure, and not harmful to the environment
- Catalysts in nanoparticle form could be extremely efficient in terms of mass of catalyst required

Homogeneous vs Heterogeneous Catalysts

- **Homogeneous:** Same phase as reactants and products
 - Used in reactions, but reformed at the end
- **Heterogeneous:** Different phase to 1+ reactants/products
 - Reactants adsorb to catalyst surface, bonds disrupted and more easily rearranged
- Heterogeneous catalysts are preferred as it is easier to separate products at the end and easier to reuse the catalyst

Enzymes

- **Enzyme:** A biological catalyst with active sites (specifically shaped sections) which can interact with a substrate (specific reactant)
- Can increase reaction rate by up to 10^{10} times

Examples

Name	Catalyst	Type	Use/Reactions
Catalytic Converter	Platinum/Palladium/Rhodium	Heterogeneous	Cars: toxic gases into harmless gases $2\text{CO}_{(g)} + \text{O}_{2(g)} \rightarrow 2\text{CO}_{2(g)}$ $2\text{NO}_{(g)} \rightarrow \text{N}_{2(g)} + \text{O}_{2(g)}$ $2\text{C}_8\text{H}_{18(g)} + 25\text{O}_{2(g)} \rightarrow 16\text{CO}_{2(g)} + 18\text{H}_2\text{O}_{(g)}$
Haber Process	Iron	Heterogeneous	Ammonia for fertilizer, explosives $\text{N}_{2(g)} + 3\text{H}_{2(g)} \rightarrow 2\text{NH}_{3(g)}$
Hardening of fats	Nickel	Heterogeneous	Margarine $\text{C}_2\text{H}_4 + \text{H}_2 \rightarrow \text{C}_2\text{H}_6$
Esterification	H^+	Homogeneous	Esters smell nice - perfumes/scents
Destruction of Ozone	Cl	Homogeneous	$2\text{O}_{3(g)} \rightarrow 3\text{O}_{2(g)}$

Class

Friday, 28 August 2020 10:48 AM

Introduction

- $Rate = \frac{[Reactant]}{time}$
 - $[] = concentration$
- E.g. $Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}$

Time (s)	[HCl] (mol L ⁻¹)
10	50
20	40
30	32
40	26
50	22
60	22 (reaction complete)
70	22

Measuring Rates

- $HCl + NaOH \rightarrow NaCl + H_2O$
 - Neutralisation: Measure pH over time
- $Mg + 2HCl \rightarrow MgCl_2 + H_2$
 - Measure volume of gas produced
 - Change in mass
- $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
 - Measure mass of precipitate formed

Rates Booklet

Collision Theory

- Increasing the rate of reaction may be achieved by:
 1. Increasing temperature
 2. Increasing surface area (solid), concentration (solution) or pressure (gas)
 3. Adding a catalyst
- **Successful collision:** Particles must collide with the correct orientation and with a minimum amount of energy to react
 - **Activation energy (E_A):** Minimum energy required for a reaction to occur

Maxwell-Boltzmann Distribution

- Energy of particles (x) vs number of particles (y)

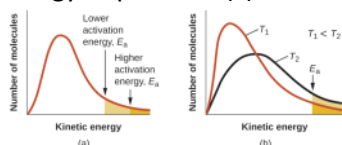


Figure 4. (a) As the activation energy of a reaction decreases, the number of molecules with at least this much energy increases, as shown by the shaded areas. (b) At a higher temperature, T_2 , more molecules have kinetic energies greater than E_a , as shown by the yellow shaded areas.

- Area under curve = total number of particles
 - No matter the temperature, the area under the curve won't change
- No molecules have no energy
- No maximum energy (should never touch the x-axis to the right)
- Only particles to the right of the activation energy have enough energy to react

Temperature

- Increased temperature = more particles with energy $\geq$ activation energy

- Greater chance of successful collisions
- Reaction rate doubles for a 10°K/C rise in temperature

Effect on Maxwell-Boltzmann Distribution - (b) in diagram above

- The peak is:
 - Later
 - Lower
 - Flatter
- More particles with $E \geq E_a$

Concentration/Pressure/Surface Area

- Increased concentration/pressure = more particles in given space
 - More collisions, more successful collisions
- If a reactant is solid, reactions may only take place on its surface
 - Smaller sized solid particles = greater area for collisions to take place = more collisions = more successful collisions

Catalysts

- **Catalyst:** Substance which speeds up the rate of reaction and is chemically unchanged by the reaction
 - Provides an alternative pathway with a lower activation energy for the reaction
- Show using a dotted line in enthalpy profile diagrams
- Lowers E_a in Maxwell-Boltzmann Distribution (further left)

Homogeneous vs Heterogeneous

- Homogeneous: Same phase as reactants and products
 - Used in reactions, but reformed at the end
- Heterogeneous: Different phase to 1+ reactants/products
 - Used as a site where reactants can adsorb to for reactions to occur

Steps (Extension)

1. **Adsorption:** of reactant to catalyst surface
2. **Reaction:** Bonds in reactants weaken at surface of catalyst, lowering activation energy
3. **Desorption:** of products from surface of catalyst; New bonds form between reactants forming the products

Examples

Reaction	Catalyst	Type	Use
Catalytic Converter	Platinum/ Rhodium	Heterogeneous	Cars: toxic gases into harmless gases CO, NO _x , C _n H _y into CO ₂ , H ₂ O, N ₂
Haber Process (N ₂ + 3H ₂ → 2NH ₃)	Iron	Heterogeneous	Ammonia for fertilizer, explosives
Hardening of fats (C ₂ H ₄ + H ₂ → C ₂ H ₆)	Nickel	Heterogeneous	Margarine
Esterification	H ⁺	Homogeneous	Esters smell nice - perfumes/scents
Destruction of Ozone (2O ₃ → 3O ₂)	Cl	Homogeneous	

14.1 - Introducing Rate

- Rate increases with temperature, concentration, pressure and state of subdivision
- Catalysts will also increase the rate of reaction
 - Catalyst: Unique substances involved in a reaction with no net consumption

14.2 - Measuring Reaction Rate

- Speed of reaction described by rate of formation of products/rate of consumption of reactants
- $Rate = \frac{\text{Amount of substance used/produced}}{\text{Time taken}}$
 - Many different units:
 - $mol\ s^{-1}$
 - $g\ s^{-1}$
 - $mL\ s^{-1}$
- Graph the amount of substance used/produced over time
 - Rate = gradient of the slope
- Over time, the rate will decrease to 0 as all the reactants are consumed

14.3 - Collision Theory

- Collision theory states that for a reaction to occur:
 - Particles must collide
 - The collision energy must be $\geq$ activation energy (E_a)
 - They must collide with the right orientation
- If all conditions are met, the particles will form a transition state/activated complex
- Transition state/Activated complex: Unstable arrangement of atoms in which reactant bonds are breaking and new bonds are forming

14.4 - Potential Energy Profile (Enthalpy Profile Diagram)

- As particles approach each other, repulsion between electron clouds causes them to slow down and lose kinetic energy
- E_k is converted into CPE
- If particles have enough E_k to collide they can form an activated complex
- Enthalpy Profile Diagrams show the change in CPE of reacting particles

14.5 - Collision Energy and Temperature

- Temperature: Measure of the average kinetic energy ($E_k = \frac{1}{2}mv^2$) of the particles of a substance
- Higher temperature = more collisions + higher average collision energy = faster rate

14.6 - Understanding the Factors that Affect Reaction Rate

- $\uparrow$ Concentration/ $\uparrow$ Pressure = $\uparrow$ Collisions = $\uparrow$ Successful collisions = $\uparrow$ Rate
- $\uparrow$ Temperature = $\uparrow E_k$ = $\uparrow$ Collisions + $\uparrow$ Collisions with energy $\geq E_a$ = $\uparrow$ Rate
- Subdivision:
 - In heterogeneous reactions (reactants in different phases/two immiscible liquids), particles can only collide at the surface of reactants
 - $\uparrow$ Surface area = $\uparrow$ Collisions = $\uparrow$ Rate
- Catalysts:
 - No net consumption of the catalyst at the end of the reaction
 - Many transition metals are used as catalysts

- Provides an alternate reaction pathway with a lower activation energy

14.7 - Enzymes: Biological Catalysts

- Inorganic Catalyst: Inorganic elements/compounds used as catalysts
 - E.g. Transition metals, simple ionic compounds
- Enzymes:
 - Very specific and fast
 - Specific shape allows substrate (reactant molecules) to fit perfectly on its surface

14.8 - Catalysts Contributing to Sustainability

- Synthesis of many chemicals needs to be sped up for economic reasons
- Increasing temperature and pressure requires lots of energy, expensive, and to the environment
- Catalysts are the solution in many cases
- Haber-Bosch Process: $\text{N}_{2(g)} + 3\text{H}_{2(g)} \rightleftharpoons 2\text{NH}_{3(g)}$
 - Iron (*Fe*) catalyst
 - Production of ammonia for fertilisers
- Catalytic Converters (Cars): Converts toxic gases into harmless non-toxic substances
 - Platinum/Palladium and Rhodium
 - Very high surface area: Fine particles spread over the surface of a supporting structure (ceramic honeycomb monolith)
 - Converts NO_x to N_2 , CO and C_xH_y to CO_2 and H_2O
- Heterogeneous Catalyst: Catalyst and reagents in different phases (gas vs solid)
 - Reagents adsorb to catalyst surface, bonds more easily rearranged hence lower E_a

14.9 - Nanoform Catalysts

- Nanomaterials have very large surface area to volume ratio
- Catalysts in nanoparticle form could be extremely efficient in terms of the mass of catalyst required

Chapter 18 - Rates of Chemical Reactions

18.1 - Investigating the Rate of Chemical Reactions

Factors that Affect Reaction Rates

- **Surface Area (Solids):** Smaller particles have larger surface area than the same mass of a larger particle
- **Concentration (Solutions):** More particles closer together → More collisions
- **Pressure (Gases):** Particles closer together → More collisions
- **Temperature:** Higher temp = faster reaction
- **Catalyst:** Allows the reaction to follow a more energetically favourable pathway

Measuring Rates Experimentally

- Must measure how much of a reactant is being used up or how much of a product is forming
- May measure using:
 - Mass
 - Volume
 - Colour change
 - pH

18.2 - Collision Theory

- Collision theory states that for a reaction to occur, particles must:
 - Collide with sufficient energy to break reactant bonds, and
 - Collide with the correct orientation to allow reactant bonds to break and product bonds to form
- **Activation energy (E_a):** Minimum collision energy for a reaction to occur
 - Higher E_a means slower reaction rate as less collisions are successful
- **Transition State:** Unstable arrangement of atoms with maximum potential energy where reactant bonds are breaking and product bonds are forming

18.3 - Applying Collision Theory

- To increase collision frequency, increase concentration/pressure/surface area
 - Make sure to discuss effect on collision frequency and number of successful collisions
- To increase collision energy, increase temperature
 - Increases frequency of collisions and energy of collisions, more collisions have energy $\geq E_a$

Maxwell-Boltzmann Distribution

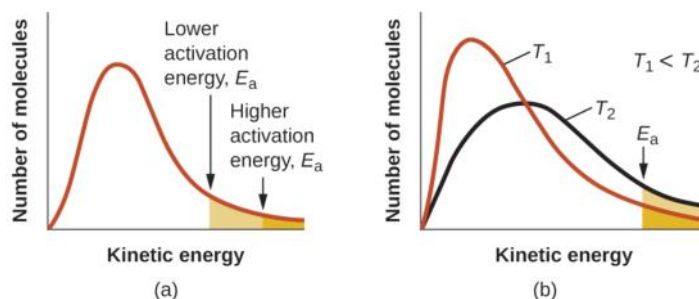


Figure 4. (a) As the activation energy of a reaction decreases, the number of molecules with at least this much energy increases, as shown by the shaded areas. (b) At a higher temperature, T_2 , more molecules have kinetic energies greater than E_a , as shown by the yellow shaded area.

- Area under curve = total number of particles
- Never touches x-axis on right
- $EK = \frac{1}{2}mv^2$
 - As temperature rises, average kinetic energy also rises, more collisions have energy $\geq E_a$ → more successful collisions

- Increasing temperature by 10°C will approximately double reaction rate
- Total area under curve won't change

Catalysts

- Provides an alternate reaction pathway with lower E_a , shifts E_a left (Diagram 1)

Chapter 19 - Catalysts

Catalysts and Activation Energy

- Catalysts provide an **alternative reaction pathway** with a lower activation energy
 - Note: The original reaction pathway still exists - not 'substituted'
- Greater proportion of collisions have collision energy $\geq E_a \rightarrow$ More successful collisions
- Can be shown on both the Enthalpy Profile Diagram and the Maxwell-Boltzmann Distribution

Types of Catalysts:

- **Homogeneous Catalysts:** Same phase as the reactants and products
 - E.g. Chlorine catalyses the destruction of ozone into O_2
- **Heterogeneous Catalysts:** Different phase to reactants and/or products

Haber Process

- $N_2 + 3H_2 \rightarrow 2NH_3$, Fe catalyst
- N_2 and H_2 adsorb to the iron, and their covalent bonds break
- In this state, they readily combine to form ammonia and are released from the iron
- Without catalyst: $> 3000^\circ C$ required
- With catalyst: $500^\circ C$ required

Catalysts in Industry

- Heterogeneous catalysts are preferable because they are:
 - Easier to separate from the products
 - Easier to reuse
 - Able to be used at high temperatures
- Adsorption of gas particles onto solid catalysts distorts bonds and can potentially even break them, allowing the reaction to proceed more easily
- Powdered/Sponge-like form used to provide the greatest surface area, allowing more molecules to adsorb
- Allows products to be made at the same rate but with much lower temperatures (and therefore energy costs)

Enzymes

- **Enzyme:** A protein with active sites (specifically shaped sections) which can interact with a substrate (specific reactant)
 - Biological catalyst
- Very efficient - Can increase reaction rates by up to 10^{10} times
- Uses:
 - Detergents (remove stains)
 - Soften leather
 - Convert lactose into lactic acid for yoghurt
 - May be used to make biofuels in the future

Enzymes vs Inorganic Catalysts

- Enzymes:
 - Produce much faster reaction rates
 - More specific (only catalyses one reaction)
 - Much more sensitive to pH/temperature changes (shape changes)

Nanoparticles as Catalysts

- High surface area to volume ratio makes them a potentially ideal catalyst
- One car manufacture has already announced they will be using nanocatalysts for more efficient pollution control which will use 70% less metal

Catalytic Converters in Cars

- Cleans exhaust gases from the engine and reduces air pollution
- Many reactions happen, including
 - $2\text{NO}_{(g)} \rightarrow \text{N}_{2(g)} + \text{O}_{2(g)}$
 - $2\text{CO}_{(g)} + \text{O}_{2(g)} \rightarrow 2\text{CO}_{2(g)}$
 - $2\text{C}_8\text{H}_{18(g)} + 25\text{O}_{2(g)} \rightarrow 16\text{CO}_{2(g)} + 18\text{H}_2\text{O}_{(g)}$
- The catalyst is a mixture of platinum, palladium, rhodium and aluminium oxide mounted on a honeycomb-shaped structure with millions of tiny pores to provide a large surface area
- Could not be used until unleaded petrol was available, as lead coated the catalyst's surface, rendering it useless

CAP 5

Sunday, 13 September 2020

8:10 AM

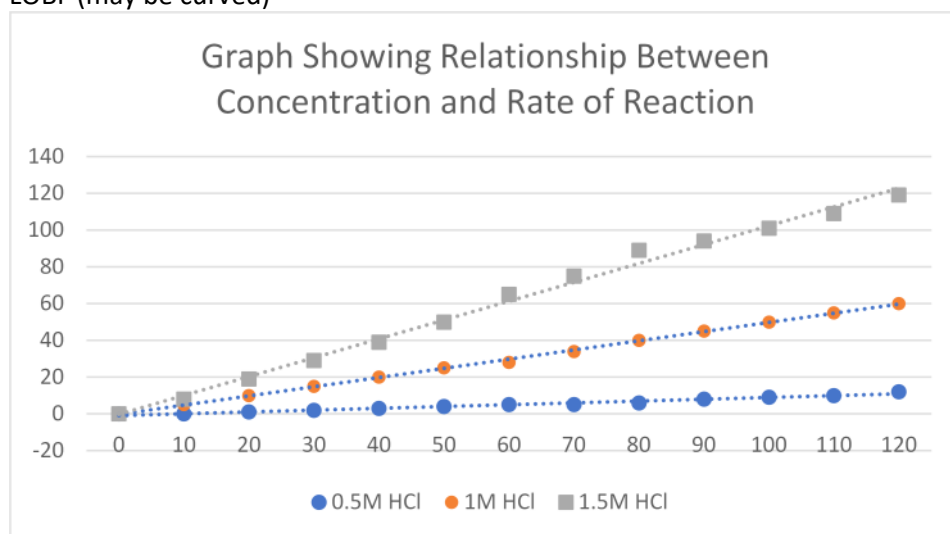
Rates of Reaction

Temperature

1. Hypothesis: *If the temperature of the reactants increases, then the rate of reaction will increase*
2. Variables
 - a. Independent: *Temperature*
 - b. Dependent: *Rate of reaction*
 - c. Controlled: *Concentration of reactants, agitation, volume of solution, shape of container used*
3. Table of Results

Time (s)	0.5M HCl	1M HCl	1.5M HCl
0	0	0	0
10	0	5	8
20	1	10	19
30	2	15	29
40	3	20	39
50	4	25	50
60	5	28	65
70	5	34	75
80	6	40	89
90	8	45	94
100	9	50	101
110	10	55	109
120	12	60	119

4. Graph
 - a. LOBF (may be curved)



5. Conclusion
 - Support/Doesn't support hypothesis
 - Use data to back up conclusion

The data supported the hypothesis that if the temperature of reactants increases, the rate of reaction will increase. At 20°C, it took 35.05 seconds for a certain amount of reaction to occur. At 60°C, it only took 5.7 seconds for the same amount of reactions to occur.

At higher temperatures, particles move faster, having both higher kinetic energy and more collisions. More collisions mean there will be more successful collisions, and the particles also have more energy which shifts the Maxwell-Boltzmann Distribution to the right

6. Evaluation

- Comment on errors and uncertainty
- How would you increase reliability and accuracy of the experiment?

Volume of solutions would not be precise ($\pm 1\text{mL}$). To solve this problem, a volumetric pipette could be used to measure the volume instead.

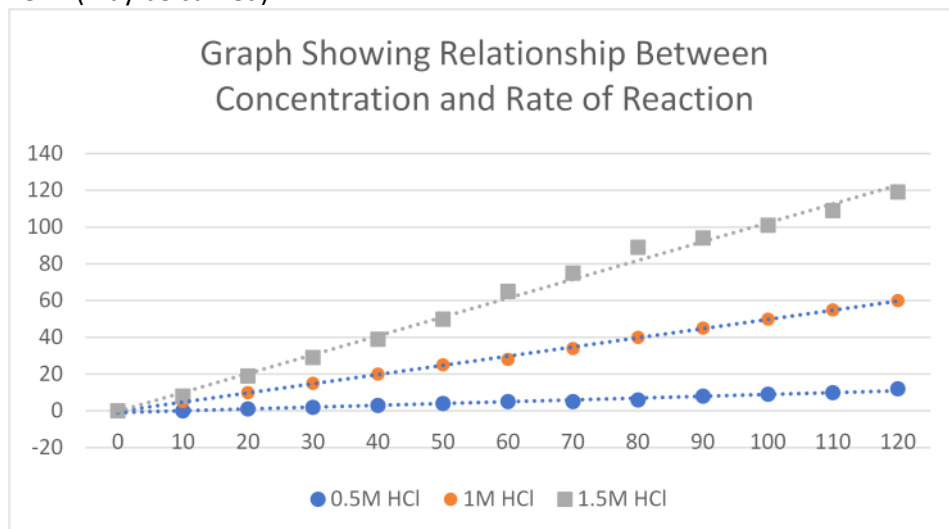
Concentration

1. Hypothesis: If the concentration of HCl increases, then the rate of reaction will increase and hydrogen gas will be collected in the test tube faster
2. Variables
 - a. Independent: Concentration of HCl
 - b. Dependent: Rate of reaction
 - c. Controlled: Volume of HCl, Mass of CaCO_3 , Shape of flask, Measuring cylinder, Temperature
3. Table of Results

Initial Temperature	Time for Cross to Disappear
20	35.05
30	20.2
40	9.25
50	7.5
60	5.7

4. Graph

- a. LOBF (may be curved)



5. Conclusion

- Support/Doesn't support hypothesis
- Use data to back up conclusion

The data supports the hypothesis, which stated that if the concentration of HCl increases then the rate of reaction will increase. This is evident as with a 0.5M HCl solution, after 60 seconds there was only 5mL of H_2 collected, whereas with the 1.5M HCl solution, after 60 seconds 65mL of H_2 was collected. This is because with a higher concentration, there are more particles in a given space, so more

collisions will occur between them. More collisions means there will be more successful collisions, and therefore the rate of reaction will be higher.

6. Evaluation

- Comment on errors and uncertainty
- How would you increase reliability and accuracy of the experiment?

The volume of HCl was not precise ($\pm 1\text{mL}$), a volumetric pipette should have been used instead.

The CaCO_3 used was not uniformly shaped, and therefore would have different surface areas in each trial. Increased surface area would have led to a faster rate of reaction, invalidating the results.

The volume of gas measured was also not accurate as it was done in a large measuring cylinder. A gas syringe could be used instead to increase the precision of the measurement.